Thermodynamic description of the Ag–Sn system: a synthetic benchmark document

ORCHID synthetic benchmark (generated; no human authors)

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Abstract

This synthetic document describes the Ag–Sn binary system as calculated with pycalphad from the SGTE dataset AgSn-12Gie.tdb; every figure, table and stated value is generated, and the exact values are kept in a separate truth key. The highest invariant reaction, (Ag) + L → ζ, lies at 719 °C.

1. Introduction

The Ag–Sn system belongs to the 20 binaries of the ORCHID benchmark, chosen to cover diagram types such as peritectic. The thermodynamic description is the SGTE collection entry [1], which reproduces the assessment of W. Gierlotka, J. Electron. Mater., 41, 86-108(2012). [2]. The calculations use pycalphad 0.11.2 at 1 atm.

For Ag, the liquidus meets the axis at 962 °C. For Sn, the liquidus meets the axis at 232 °C.

2. Phase diagram and invariant reactions

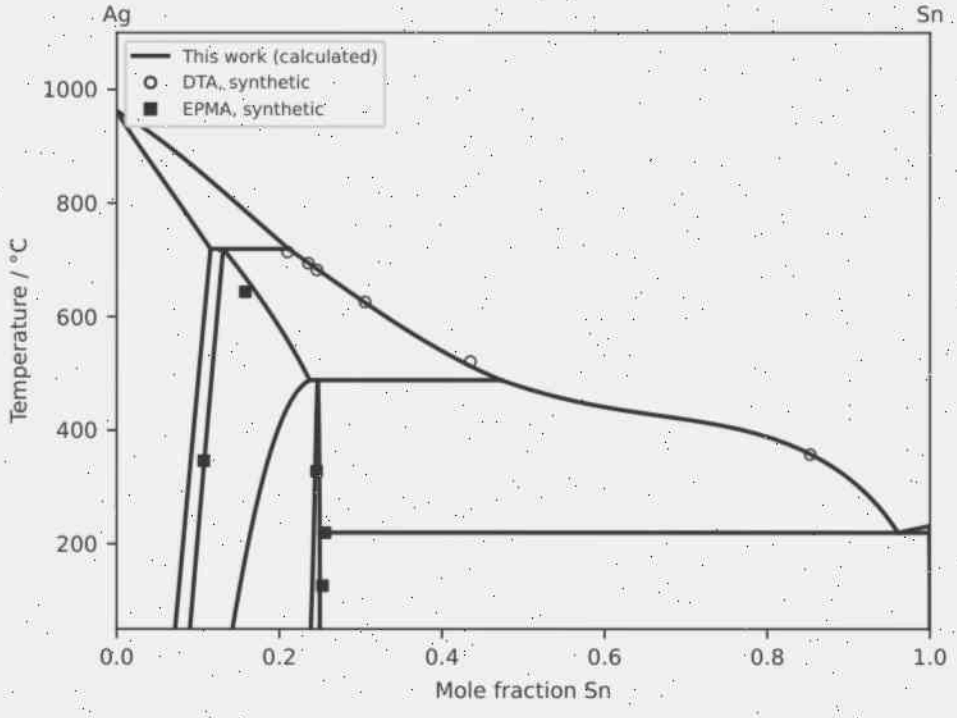


Fig. 1. Calculated Ag–Sn phase diagram with synthetic experimental-style points.

We obtain 719 °C for the peritectic reaction ((Ag) + L → ζ), where L has x(Sn) = 0.211. The dataset places a peritectic (ζ + L → Ag3Sn) at 488 °C; the composition of L is a mole fraction of 0.473. The eutectic reaction L → Ag3Sn + (Sn) is calculated at 220 °C, with L at x(Sn) = 0.961.

Table 1 compares the calculated temperatures with experimental values from the literature [3–5]; in this synthetic document those values and references are fictitious.

Table 1. Invariant reactions and congruent points, calculated and experimental (Ag–Sn).

Reaction	Type	T / °C		Compositions (x(Sn))		
		this work	exp.	phase 1	phase 2	phase 3
(Ag) + L → ζ	peritectic	719.3	722.3 [4]	0.115	0.211	0.131
ζ + L → Ag3Sn	peritectic	488.2	492.2 [5]	0.237	0.473	0.247
L → Ag3Sn + (Sn)	eutectic	219.8	–	0.961	0.250	0.999

Experimental temperatures are fictitious values attributed to invented references [3–5] for testing evidence-class extraction; they are not data.

3. Phase stability

Table 2. Solid phases: structure, stability range and homogeneity range (Ag–Sn).

	Structure			Stability range / K		Homogeneity range	
Phase	Pearson	Strukturb.	Prototype	from	to	at.% Sn	at T / K
(Ag)	cF4	A1	Cu	< 300	1234.9	0.0–11.5	990
ζ	hP2	A3	Mg	< 300	992.4	11.6–23.5	760
Ag3Sn	oP8	D0a	β-Cu3Ti	< 300	761.4	23.8–25.0	300
(Sn)	tI4	A5	β-Sn	< 300	505.1	99.9–100.0	490

Structure labels follow the TDB phase names (– where the name does not identify a structure). Stability range: temperatures between which the phase is stable in some part of the diagram.

According to Table 2, Ag3Sn remains stable up to 488 °C (peritectic). The solubility of Sn in (Ag) reaches a mole fraction of 0.115 at 719 °C.

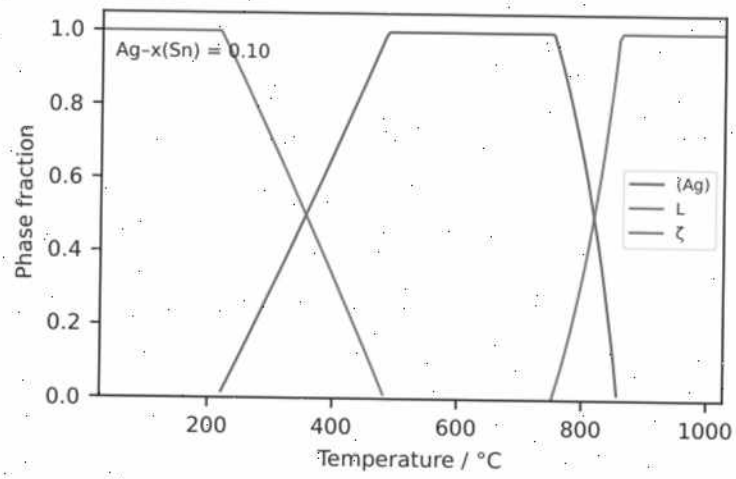


Fig. 2. Calculated equilibrium phase fractions versus temperature for Ag–x(Sn) = 0.10.

4. Thermodynamic properties

At 1300 K the integral mixing enthalpy of the liquid has its minimum, −3.15 kJ/mol, at x(Sn) = 0.220. The calculated enthalpy of formation of Ag3Sn at 298.15 K is −3.95 kJ/mol per mole of atoms. Per formula unit Ag3Sn this is −15.8 kJ/mol. Its heat capacity at room temperature is 25.77 J/(mol K). For the equiatomic liquid at 1300 K, the Gibbs energy of mixing is −10622 J/mol.

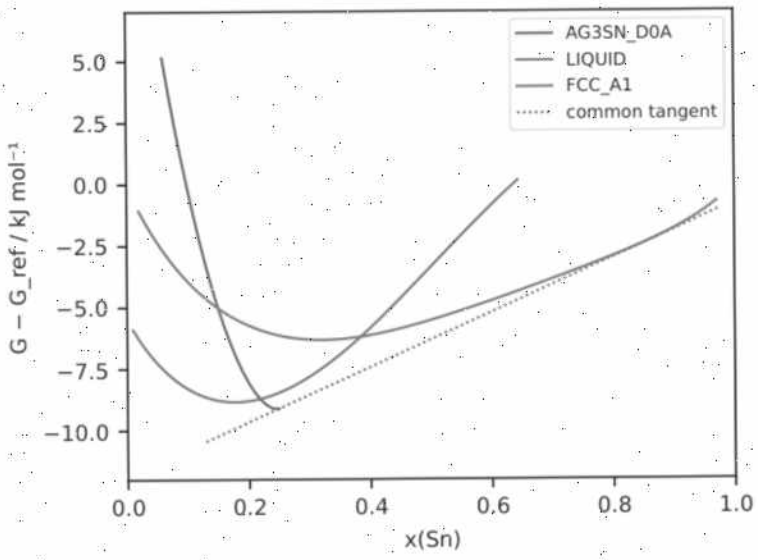


Fig. 3. Calculated Gibbs energy curves at 630 K.

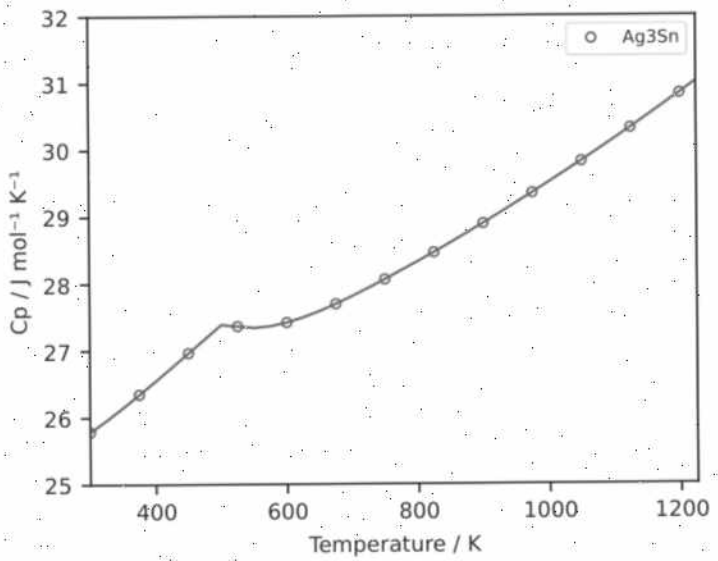


Fig. 4. Calculated heat capacity of Ag3Sn.

Table 3. Calculated formation properties of the intermediate phases at 298.15 K (Ag–Sn).

Phase	at.% Sn	ΔfH(298.15 K) / kJ mol⁻¹	ΔfS / J mol⁻¹ K⁻¹	Cp(298.15 K) / J mol⁻¹ K⁻¹
Ag3Sn	24.4	−3.95	2.40	25.77
ζ	15.7	−2.67	3.21	25.64

Reference: stable elements (SER) at 298.15 K; values per mole of atoms.

Table 4. Integral and partial mixing properties of the liquid at 1300 K (Ag–Sn).

	Liquid at 1300 K			
at.% Sn	ΔmixH / J mol⁻¹	ΔmixG / J mol⁻¹	a(Ag)	a(Sn)
10	−2395	−5701	0.8527	0.0215
20	−3139	−8725	0.6706	0.0874
30	−2853	−10292	0.5093	0.2019
40	−2044	−10834	0.3850	0.3416
50	−1101	−10622	0.2935	0.4773
60	−301	−9825	0.2244	0.5952
70	196	−8526	0.1669	0.6980
80	346	−6706	0.1121	0.7958
90	219	−4187	0.0553	0.8969

5. Conclusions

This document is synthetic and exists only to test extraction tools; its truth key lists every boundary, field, invariant, plotted point, table cell and stated value.

References

[1] B. Hallstedt, SGTE binary dataset collection, Calphad 89 (2025) 102833 (CC BY 4.0).

[2] W. Gierlotka, J. Electron. Mater., 41, 86-108(2012).

[3] K. Synthetica, L. Exemplar, J. Fict. Phase Equilib. 12 (1987) 345–352 — fictitious reference, invented for this synthetic benchmark.

[4] M. Placeholder, Z. Notreal, Fict. Thermochim. Acta 41 (1994) 77–89 — fictitious reference, invented for this synthetic benchmark.

[5] R. Imaginary, Proc. Synth. Calorim. Soc. 8 (2003) 1102–1110 — fictitious reference, invented for this synthetic benchmark.